# EventRoams Plugin

Map Marker & HUD Performance Audit

Generated 10 November 2026

## Summary

The map marker and the HUD are already well built. No large allocation problems and no per-player text building were found: the marker name and the HUD status text are already compared on their inputs or built once per language, the HUD uses partial refresh with shared template values, collections are pooled and freed, and the scoreboard rebuild is coalesced to at most once per 5 s. The remaining issues are a few paths a player can trigger on purpose, and some sends that repeat without need.

| # | Severity | Finding | Status |
|---|---|---|---|
| 1 | Medium | Repeated crossing of the zone border runs the full HUD / enter path each time | Verified in code; rate estimated |
| 2 | Low-Medium | EnterPlayer calls ServerPanel before the already-inside check | Verified in code |
| 3 | Low-Medium | Radius marker re-sent every 10 s although nothing in it changes | Unverified (needs decompile) |
| 4 | Low | HUD re-sent to every viewer every 5 s even when nothing changed | Verified in code |
| 5 | Low | BasePlayer.FindByID used in several loops | Unverified (needs decompile) |

## Scope and limits

- Read: about 900 of the 7,039 lines. That covers the map marker, the HUD controller, both zones, the scoreboard rebuild path and the image helpers.
- **Verified** means the plugin code was read. Anything about vanilla Rust or Oxide behaviour that could not be decompiled in this session is marked **unverified** and is not stated as fact.
- The CUI send and destroy cost itself belongs to the cui-partial-refresh-audit skill, which was not available. It is not audited here; this report only covers how often the plugin sends.
- The plugin's AGENTS.md and the game and Oxide DLLs were not available. Nothing was compiled or measured on a server. Call rates are estimates.
- No code was changed. The fixes below are proposals; findings 1, 2 and 4 change timing or behaviour and need approval.

## Findings

**1** **MEDIUM** **Crossing the zone border over and over runs the whole HUD path each time**

A player can cross the GUI zone edge as fast as they can strafe. Each crossing does the following (lines refer to EventRoams.cs):

- **Enter HUD zone (:3313, :5395):** UpdateFields() loops over everyone in the zone for the team count (:5164). Then ShowFull runs: a Clans plugin call, a full HUD build and send with per-player element clones, TryResetPlayerSkins and the OnEventEnter hook into other plugins.
- **At night:** SwitchTimeForPlayer toggles the admin flag and calls SendNetworkUpdateImmediate() twice, plus console commands (:3338, :3342). The cost of that player update is unverified.

- **Enter event zone (:3256):** a chat or toast message, another Clans call and crafting.CancelAll().
- **Leave (:5415):** DestroyUi, which also fires the OnDestroyUI hook into other plugins.

The cost grows with the number of players in the zone, because of the team count. With 20 players jittering on the edge at a couple of crossings per second, that is dozens of full HUD sends a second (estimate).

**Proposed fix (needs approval):** only run the HUD part of the enter path if the last full show for that player was more than 3 s ago. GuiCoroutine already sends the full HUD to anyone in the zone who has no mask, so a skipped player gets it within one 5 s pass. Players see no difference apart from a short delay when they re-enter quickly.

```
private const float EnterShowCooldown = 3f;
private Dictionary<ulong, float> _lastFullShow =
    Pool.Get<Dictionary<ulong, float>>();   // free in Delete()

public void OnPlayerEnterToGuiZone(BasePlayer player)
{
    if (_deleted) return;

    float now = Time.realtimeSinceStartup;
    if (_lastFullShow.TryGetValue(player.userID, out float last) && now - last < EnterShowCooldown)
        return;                       // GuiCoroutine sends the HUD on its next pass

    _lastFullShow[player.userID] = now;
    UpdateFields();
    ShowFull(player);
}
```

The same cooldown idea can be applied to the event-zone enter message and the night time switch.

| 2 | LOW-MEDIUM | EnterPlayer calls ServerPanel before the already-inside check |
|---|---|---|

Line :3698 runs DeleteOtherPluginsUi(player), a plugin call, before the PlayersInZone.Add check at :3700. OnTriggerEnter finds the player with GetComponentInParent<BasePlayer>. If it fires once per collider of the player, every entry makes several of these calls. How often the trigger fires per player was not verified.

**Proposed fix**, if the repeated panel close is not intended:

```
if (!PlayersInZone.Add(player.userID)) return;
_ins.DeleteOtherPluginsUi(player);
OnPlayerEnterZone(player);
```

| 3 | LOW-MEDIUM | Radius marker re-sent every 10 s although nothing in it changes |
|---|---|---|

Lines :1534-1540 call UpdateMapMarker() every 10 s. That calls SendUpdate() and then SendNetworkUpdate() on the radius marker (:3145-3146). Its radius, colours and alpha are set once at spawn (:3097-3100). The marker is global, so each send goes to every connected player. PlayerLogin already sends the marker to anyone who joins (:252).

**Unverified:** SendUpdate() may already include a network update, which would make the second call a duplicate. The decompiled MapMarkerGenericRadius.SendUpdate is needed to confirm. If the periodic send exists only to catch late joiners, it can be removed or stretched to about 60 s.

The vending marker is fine: it only sends when the minute shown in its name changes (:3156-3162).

| 4 | LOW | HUD re-sent to every viewer every 5 s even when nothing changed |
|---|---|---|

The time label and the status countdown already tick on the client. The progress bar moves less than one unit per pass (a 1,800 s event across a 200-unit bar is about 0.55 units per 5 s). The counters rarely change. Most of the traffic is therefore resync.

**Proposed fix (needs approval):** in GuiCoroutine, skip the per-player refresh unless something changed: a counter value, a crate-state flip, a top rebuild, or the bar moving by 1 unit or more. Resync the timer every 30-60 s regardless.

This only changes how often sends happen. Their per-send cost is for the CUI audit.

| | | |
|---|---|---|
| 5 | LOW | **BasePlayer.FindByID used in several loops** |

It is called per player in GuiCoroutine (:5213), per solo scorer in GetSortedTop (:2876) and in RemovePlayersFromZone (:3723). If it scans activePlayerList, the cost is players times active players every 5-10 s: tens of thousands of comparisons, which is negligible today. This is unverified. If it ever matters, store BasePlayer references and skip the lookup.

## Checked and left alone

- **GetMarkerNameKey:** compares the whole-minute input, not the built string (Pattern 3 already done).
- **Status text and dot position:** built once per language per pass (Pattern 2 already done).
- **HUD updates:** pooled collections are freed in Delete; partial refresh with per-group containers; destroys go only to confirmed viewers; shared values are written on the templates without per-player clones; the scoreboard rebuild is coalesced to at most once per 5 s.
- **Image lookups:** misses are cached for 10 s, so an unresolved URL costs two plugin calls per window.
- **Zone checks:** IsPositionInZone uses cached copies of the position and radius.
- **Event-zone dome spheres:** spawned once. Their count comes from the config (Darkening and Brightness), so keep those low.

## Open questions for the dev

- Apply fixes 1, 2 and 4? The plugin needs to be in the repo, and it has to be compiled and tested on your side.
- Is the repeated ServerPanel close in finding 2 intentional?
- Please run `ilspycmd -t MapMarkerGenericRadius` and `ilspycmd -t BasePlayer` to settle findings 3 and 5.
- Roughly how many players are usually in the zone?